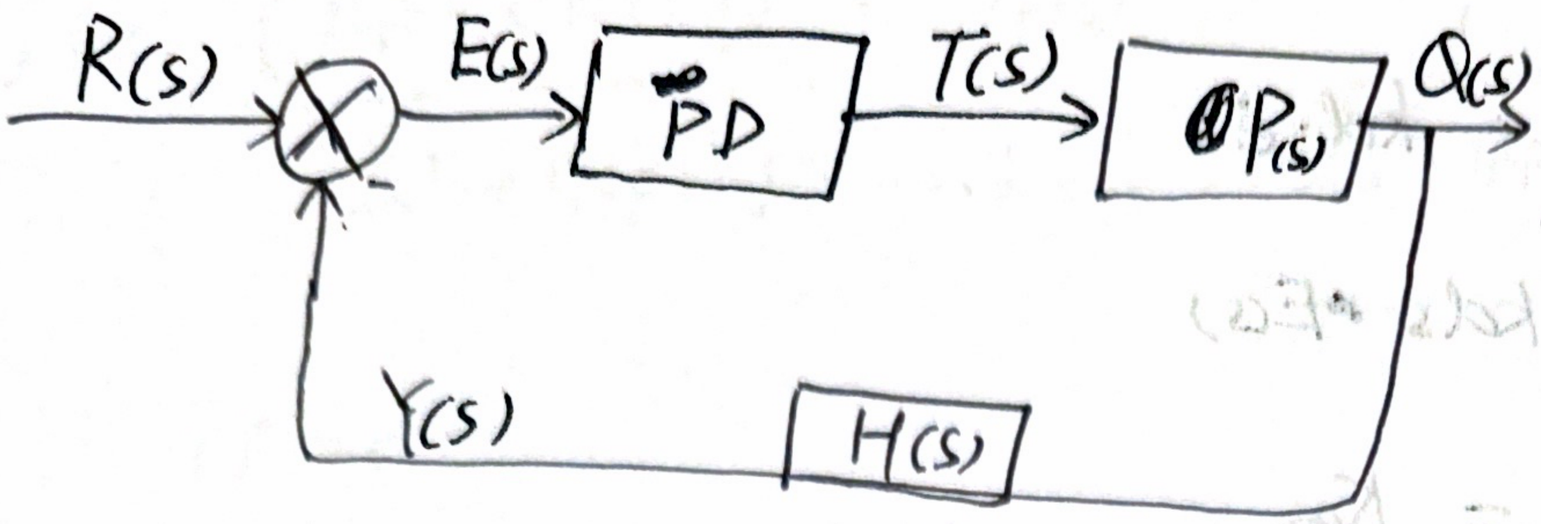


$$Q(s) = E(s) \cdot K_{PD} \cdot P(s)$$

$$Y(s) = Q(s) \cdot H(s)$$

$$E(s) = -Y(s) + R(s) = R(s) - QH = R - QH = E.$$

$$T = K_{PD} \cdot E = K_{PD} \cdot (R - QH).$$

$$Q = P \cdot T = P \cdot K_{PD}(R - QH) = PK_{PD}R - PK_{PD}QH$$

$$(1 + PK_{PD}H)\,Q = PK_{PD}R.$$

$$\frac{Q}{R} = \frac{PK_{PD}}{1 + PK_{PD}H} = \frac{G}{1 + GH} \rightarrow G = PK_{PD}$$

P: Physical plant

PK_{PD}: PD-Controller.

H: feedback gain.

for PD controller:

$$\tau = k_p \cdot e + k_d \cdot \dot{e}$$

$$T(s) = K_p E(s) + k_d s \cdot E(s)$$

$$\frac{T}{E} = k_p + k_d s = K_{PD}$$

for second order system:

$$I\ddot{q} + c\dot{q} + kq = \tau$$

$$(Is^2 Q - sq_0 - \dot{q}(0)) + (csQ - cq_0) + kQ = T$$

$$(Is^2 + cs + k)Q = \frac{(s+c)q_0 + \dot{q}_0 + T}{(Is^2 + cs + k)}$$

if $q_0 = \dot{q}_0 = 0$

then $$Q(s) = \frac{T(s)}{Is^2 + cs + k} \rightarrow \frac{Q}{T} = \frac{1}{Is^2 + cs + k} = P(s)$$

so

$$G(s) = PK_{PD} = \frac{k_p + k_d s}{Is^2 + cs + k}$$

if $H = 1$ (encoder feed back q_m)

then total transfer: $$1 + GH = 1 + G = \frac{(Is^2 + cs + k) + k_p + k_d s}{Is^2 + cs + k}$$

$$\frac{G}{1 + GH} = \frac{k_p + k_d s}{Is^2 + cs + k} \cdot \frac{Is^2 + cs + k}{(Is^2 + cs + k) + k_p + k_d s} = \frac{k_d + k_d s}{Is^2 + cs + k + k_p + k_d s}$$

$$\frac{Q}{R} = \frac{k_d + k_d s}{Is^2 + (c + k_d)s + (k + k_p)}$$

$$Q(s) = \frac{(kp + s\,kd)Qd + Tf(s)}{Is^2 + (kp+c)s + kp} = \frac{1}{Is^2+(kp+c)+kp}\,[Qd\ Tf]\cdot\begin{bmatrix} kp+skd \\ 1 \end{bmatrix}$$

→ std 2nd ODE: $m\ddot{x} + c\dot{x} + kx = 0$

$$\frac{-c \pm \sqrt{c^2-4mk}}{2m}$$

when $Tf = 0$:

$$Q(s) = \frac{(kp+skd)\,Qd}{Is^2 + (kp+c)s + kp}$$

$\sqrt{c^2-4mk} \geq 0$

$$= \frac{kp\,Qd}{Is^2+(kp+c)s+kp} + \frac{skd\,Qd}{Is^2+(kp+c)s+kp}$$

$c^2 \geq 4mk$

$c \geq 2\sqrt{mk}$

$\downarrow L(s)$

$(ms^2X - msX_0 - mV_0) + CsX - CX_0 + kX = 0$

$ms^2X + CsX + kX = msX_0 + mV_0 + CX_0$

$Is^2+(kp+c)s+kp$ to be stable:

$(ms^2 + Cs + k)X(s) = (ms+Cs)X_0 + mV_0$

$$= s^2 + \left(\frac{kp+c}{I}\right)s + \frac{kp}{I}$$

$(s-a)^2 + b^2 = s^2 - 2as + a^2 + b^2$

$$X(s) = \frac{(m+c)sX_0 + mV_0}{ms^2+cs+k}$$

let $\sqrt{\frac{kp+c}{I}} = 2a$

let $\frac{kp}{I} = a^2 + b^2$

$$\frac{kp+c}{I} = 2a \rightarrow \frac{kp+c}{2I} = a$$

$$s^2 + \frac{c}{m}s + \frac{k}{m}$$

so $\left(\frac{kp+c}{2I}\right)^2 + b^2 = \frac{kp}{I}$

$kp^2 + (2c-4I)kp + c^2 \geq 0$

$$\frac{kp}{I} - \frac{kp^2+2kpc+c^2}{4I^2} = b^2$$

$\omega = \sqrt{\frac{k}{m}}$, & $\delta = \frac{c}{2\sqrt{mk}}$.

$$\frac{(4I-4I)kp - kp^2 + (2c+4I)kp + c^2}{4I^2} = b^2$$

$\omega_{matched} = \sqrt{\frac{kp}{I}}$

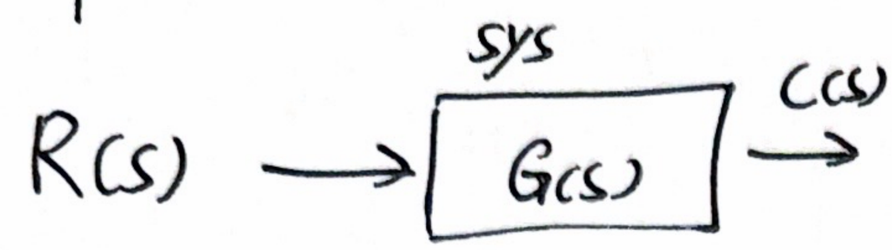
$\delta_{matched} = \frac{kp+c}{2\sqrt{I \cdot kp}}$

$0 < \delta < 1$, decay (under damp)

$\delta = 1$, decay no oscillate

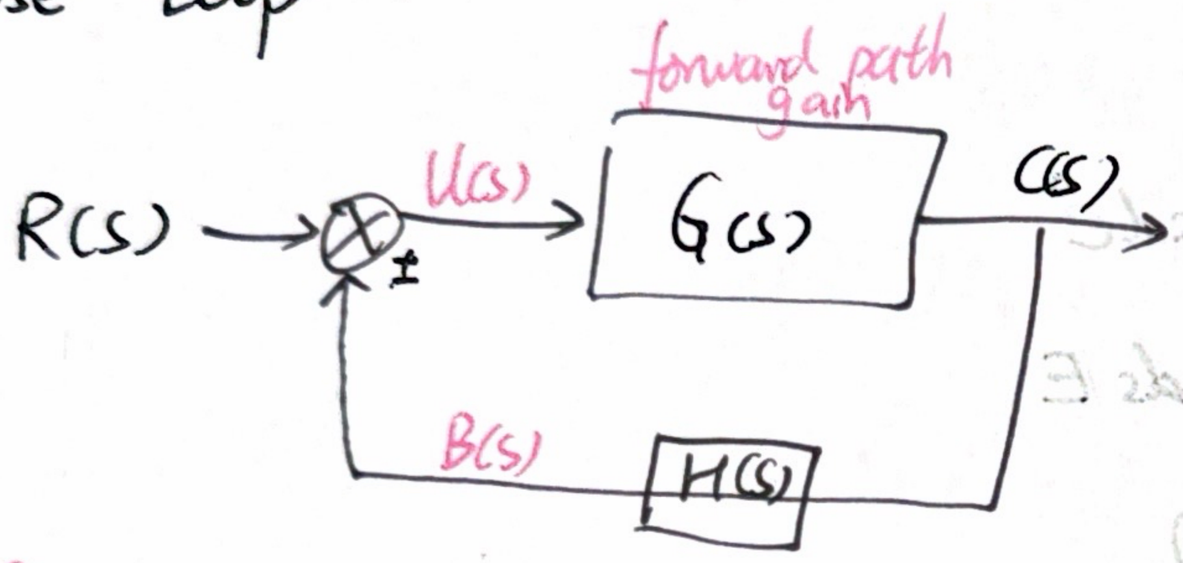
$\delta > 1$, over damped.

Open Loop :



$C(s) = G(s) R(s)$ → $G(s) = \frac{C(s)}{R(s)} = \frac{\text{output}}{\text{input}}$.

Close Loop :



Positive fdbk: encourage deviation.
negative fdbk: oppose deviation

$U(s)$ = actuating signal.

for negative feedback:

$U(s) = -B(s) + R(s)$

$B = CH$.

$U(s) = R(s) - B(s) = R(s) - C(s)H(s)$

$U = R - CH$.

$C = U \cdot G = (R - CH)G$

$C = RG - CHG$

$(1 + GH)C = RG$

$C = \frac{RG}{1 + GH}$

$\frac{C}{R}$ = transfer func = $\frac{G}{1 + GH}$.